Central-depth stability and optimal central realizations of three-Engel Lie algebras

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Abstract

Let k have characteristic two and more than two elements. We study how central depth controls the nilpotency class of ordinary three-Engel Lie algebras. For algebras with central second derived ideal and at most d generators, the sharp class maximum is $\max(5, d+2)$. The ceiling $d+2$ persists at central depths two, three and four from ranks four, five and six, at fifth depth from rank nine, and at sixth depth from rank ten. The fifth-depth start is optimal: its rank-seven and rank-eight critical layers are determinant twists of the cubic free Lie module and symmetric cube, respectively, with a nonsplit Frobenius structure in rank eight. A balanced-module peeling and suspension argument propagates exact finite starting calculations to arbitrary rank. We also determine the least center in a central realization of the free metabelian three-Engel algebra: it is $\lceil (d+1)/2 \rceil$ for $d \geq 3$, attained by a three-Engel algebra with center equal to its second derived ideal. Exact certificates accompany the finite proof inputs.

Keywords. Three-Engel Lie algebra; nilpotency class; central depth; central realization; Koszul module; cubic representation.

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1 Introduction

Let k have characteristic two and $|k| > 2$. We study ordinary Lie algebras satisfying the three-Engel identity $[y, x, x, x] = 0$. The central question is how the depth of the second derived ideal controls nilpotency class. A central lift can add a dimension without adding a class; at large generator rank this delay persists through several full layers of the universal central tower.

For $M_0(H) = H''$ and $M_{j+1}(H) = [M_j(H), H]$, the condition $H'' \subseteq Z_s(H)$ is $M_s(H) = 0$. Let $D_{d,s}$ be the relatively free object with at most d ordinary generators and this central-depth condition. We determine sharp stable ranges, the representations at the first unstable ranks, and the minimum center needed to realize the free metabelian quotient.

1.1 Sharp class boundaries and stable depths

For algebras with central second derived ideal, the exact class law is

$$e_{\mathrm{cbm}}(d) = \max(5, d + 2) \qquad (d \geq 2).$$

The complete top central cover of the free metabelian algebra H_d is controlled by $\det(U) \otimes \mathrm{Sym}^2 U$. Every top cocycle in this space is automatically three-Engel. This provides attaining objects at each exact rank and a precise first nonmetabelian boundary.

The class ceiling $d + 2$ remains sharp at central depths two, three and four for $d \geq 4, 5, 6$, respectively. It persists at fifth depth for $d \geq 9$ and sixth depth for $d \geq 10$. The fifth-depth start is optimal. Its rank-seven critical layer is $\det(U) \otimes \mathrm{Lie}^3(U)$, of dimension 112, whereas rank eight gives $\det(U) \otimes \mathrm{Sym}^3 U$, of dimension 120, with a nonsplit Frobenius submodule. The two boundary ranks therefore carry different class-raising representations, not only different numbers of parameters.

The propagation argument retains the actual balanced source module. Peeling moves a coefficient action into an outer input; full support and singleton symmetry reduce the possible critical weights. A nonlinear suspension then propagates vanishing to arbitrary rank. The finite starting systems are computer-assisted and are supplied with independent exact replays. The proof separates those finite inputs from the rank-uniform propagation, and base change proves the statements over every field in the stated range.

1.2 Optimal central realizations

For the same free metabelian H_d, $d \geq 3$, we prove

$$\min_{E/Z(E) \simeq H_d} \dim_k Z(E) = \left\lceil \frac{d+1}{2} \right\rceil.$$

The lower bound is the classical center-to-Hom injection. The sharp construction packs two-generator pair detectors and a top symmetric cocycle into exactly that number of center channels. It is itself three-Engel, has class $d + 2$, and has center equal to its second derived ideal. Minimizing the whole center differs from retaining the full kernel of a complete central cover.

1.3 Context and organization

The characteristic-two three-Engel theory and its classical generator-dependent class bounds are developed in [7, 8]. Centre-by-metabelian structures and balanced commutator modules have the antecedents [2, 1, 3]. We use their module framework to identify exact central-depth quotients, stable ranges and critical representations. The ordinary commutator model

is defined and proved within this paper; no result about quadratic orbit maps is needed for the ordinary theorems.

Quadratic orbit maps and their complete ordinary reconstruction are treated in [9]; metabelian extremal dimensions and critical type growth are treated in [10]. These are distinct questions developed from the unified research manuscript [11]. In particular the full ordinary three-Engel class here is larger than the sextic-zero category in that reconstruction.

Section 2 gives the local commutator foundation, Section 3 the top covers and sharp class law, Section 4 the stable and exceptional depths, and Section 5 the minimum-center theorem. The appendices give the complete collection, propagation and packing arguments. We use ordinary generators and dimensions over k throughout; brackets are alternating and left-normed. A nonzero abelian algebra has class one. Write $e(d)$ for the maximum class of a finite d-generated ordinary three-Engel algebra over k; its optimal growth remains open.

2 The ordinary metabelian model

Work over a characteristic-two field k with $|k| > 2$.

2.1 A multigraded commutator module

For $d \geq 2$, put

$$R_d = k[t_1, \dots, t_d]/(t_1^2, \dots, t_d^2), \qquad \mathfrak{m} = (t_1, \dots, t_d).$$

Let W_d be the R_d-module generated by v_{ij}, $i < j$, subject to

$$t_i v_{jk} + t_j v_{ik} + t_k v_{ij} = 0 \qquad (i < j < k). \tag{1}$$

We set $v_{ji} = v_{ij}$ and $v_{ii} = 0$. This is the classical free metabelian commutator-module presentation, base-changed to R_d; compare [5, Theorem 6.2] and [1, Sections 2–3].

The ordinary algebra $H_d = U \oplus W_d$, with $[x_i, x_j] = v_{ij}$, $[x_i, w] = t_i w$ and $[W_d, W_d] = 0$, is the free metabelian three-Engel algebra. In a metabelian target the derived actions commute. Partial linearization of $[y, x, x, x] = 0$ over $|k| > 2$ forces $t_x^2(H') = 0$, and Jacobi gives the triangles. Conversely these conditions imply three-Engel. Its lower-central terms are

$$\gamma_j(H_d) = \mathfrak{m}^{j-2} W_d \qquad (j \geq 2). \tag{2}$$

Grade R_d by coefficient degree, and put the generators v_{ij} in coefficient degree zero. We also use the finer grading $\deg t_i = e_i$, $\deg v_{ij} = e_i + e_j$.

Proposition 2.1. *The coefficient-degree Hilbert polynomial of W_d is*

$$H_d(T) = \sum_{k=2}^{d} \binom{d}{k} \big((k-1)T^{k-2} + kT^{k-1}\big) + \binom{d}{2} T^2. \tag{3}$$

Consequently

$$\dim_k W_d = (d-1)2^d + 1 - d + \binom{d}{2}.$$

The ordinary algebra H_d has class $\max(4, d+1)$.

Proof. A raw multihomogeneous monomial is $t_S v_{ij}$, where S is square-free. Its multidegree has entries zero, one, or two, and at most two entries are doubled. Fix its support of size $k \geq 2$.

If no index is doubled, the raw coordinates are the edges of the complete graph on the support. Relations (1) are its triangle sums. The quotient edge space has dimension $k-1$, in coefficient degree $k-2$.

If exactly one index j is doubled, only edges incident with j occur. Triangle relations through j identify every pair of these edges, leaving dimension one. There are k choices of j, in coefficient degree $k-1$.

If exactly two indices i, j are doubled, there is one raw coordinate. A third index in the support kills it by a triangle relation, since the other two terms contain a squared variable. The coordinate survives exactly when the support is $\{i, j\}$, giving the final term of (3). These cases exhaust the multigrading.

The maximum coefficient degree is two for $d = 2, 3$, and $d-1$ for $d \geq 4$. Equation (2) gives the class. □

This identifies the top layers used to study ordinary central extensions.

3 Top central covers and a sharp class boundary

We study the first nonmetabelian boundary in ordinary three-Engel theory. Put $U = k^d$ and use the classical commutator module W_d of Section 2. The ordinary Lie algebra

$$H_d = U \oplus W_d, \qquad [x_i, x_j] = v_{ij}, \quad [x_i, w] = t_i w, \quad [W_d, W_d] = 0 \tag{4}$$

is the universal metabelian 3-Engel algebra on d ordinary generators. Give x_i Lie weight one and v_{ij} weight two. For $d \geq 4$, its top weight is $d+1$.

Theorem 3.1 (The full top central cover)**.** *Let $d \geq 4$, and give a central coefficient space Z weight $d+2$. There is a natural isomorphism*

$$H^2_{\mathrm{CE}}(H_d, Z)_{d+2} \simeq \mathrm{Hom}_k\big(\det(U) \otimes \mathrm{Sym}^2 U, Z\big), \tag{5}$$

where the subscript denotes total input weight. Every class in this homogeneous space defines an ordinary 3-Engel central extension. If $\dim_k Z = 1$, every nonzero class gives an extension of exact ordinary generator rank d, class $d+2$, and second derived ideal equal to Z.

The top commutator space is naturally $\det(U) \otimes U$. The CE equations determine a cocycle from its pairing of that space with U, and the pairing is exactly a symmetric bilinear form with the displayed determinant twist. The symmetry is bilinear symmetry in characteristic two, so its diagonal coefficients remain independent.

Let $e_{\mathrm{cbm}}(d)$ be the largest class of a finite-dimensional ordinary 3-Engel k-Lie algebra of generator rank at most d satisfying $L'' \subseteq Z(L)$. Thus $L/Z(L)$ is metabelian.

Theorem 3.2 (A sharp centre-by-metabelian class law)**.** *For every $d \geq 2$,*

$$e_{\mathrm{cbm}}(d) = \max(5, d+2). \tag{6}$$

The maximum is also attained at exact ordinary generator rank d.

The upper bound is the classical metabelian ceiling $\max(4, d+1)$, obtained from W_d, followed by one central step. The top covers attain it for $d \geq 4$. A direct two-generator calculation supplies the low-rank endpoints and proves $e(2) = 5$ (Lemma A.2). Consequently any superlinear mechanism for $e(d)$ must have noncentral second derived ideal.

The commutator module and the CE/Hopf central-cover framework are classical; the free centre-by-metabelian setting is developed in [1]. Here the square-zero module relations determine the full top quotient and make its cocycles automatically 3-Engel. Appendix A gives the quotient calculation, the ordinary metabelian ceiling, and the attaining models.

4 Central depth and delayed growth of class

How many class-preserving central lifts can occur before a new class appears? The answer depends on generator rank. The sharp ceiling above persists through several larger central-action layers, with distinct low-rank endpoints.

Throughout the ordinary statements, k has characteristic two and more than two elements. Generation and the three-Engel identity refer to ordinary Lie algebras. For the relatively free three-Engel algebra $\mathcal{F}_d$ on d generators, put

$$M_0 = \mathcal{F}_d'', \qquad M_{j+1} = [M_j, \mathcal{F}_d], \qquad D_{d,s} = \mathcal{F}_d/M_s, \qquad E_d = D_{d,1}.$$

Thus $D_{d,s}$ is universal under the condition $H'' \subseteq Z_s(H)$. Here $Z_s(H)$ denotes the sth upper central term, with $Z_0(H) = 0$.

Theorem 4.1 (Sharp class ceilings at central depths two, three, and four)**.** *Let H be a finite-dimensional ordinary three-Engel k-Lie algebra generated by at most d elements. Then*

$$H'' \subseteq Z_s(H) \quad \Longrightarrow \quad \operatorname{cl} H \le d+2 \quad \begin{cases} s=2, & d \ge 4, \\ s=3, & d \ge 5, \\ s=4, & d \ge 6. \end{cases} \tag{7}$$

Every bound is attained at exact ordinary generator rank d. Equivalently, $\operatorname{cl} D_{d,s} = d+2$ in these ranges.

The second-layer bound follows from a balanced symmetric-square module. At the third layer, two polarized Engel instances and thirteen explicit depth relations give a five-variable collecting identity; nonlinear substitution propagates it to every larger rank. The fourth layer uses the next-to-top part of the same module and a separate suspension argument. Its rank-six starting step is computer-assisted: three complete multihomogeneous systems are replayed independently. The symbolic arguments and the finite starting systems are given in Appendix B.

Proposition 4.2 (Boundary ranks)**.** *The complete relatively free objects satisfy*

	$D_{d,1}$	$D_{d,2}$	$D_{d,3}$	$D_{d,4}$
$d=4$	6	6	7	*not determined here*
$d=5$	7	7	7	8

where entries are nilpotency classes. At rank five there are concrete central epimorphisms

$$A_{178} \twoheadrightarrow A_{167} \twoheadrightarrow A_{139} \twoheadrightarrow A_{118}, \tag{8}$$

with classes $8, 7, 7, 7$, respectively. Subscripts give dimensions over k; the three kernels have dimensions $11, 28, 21$. Every object has exact ordinary generator rank five. At rank four there is a corresponding $42 \to 41 \to 40$ dimensional chain of classes $7, 6, 6$.

These concrete quotients certify the lower bounds; they are not identified with the entire relatively free objects. The upper bounds follow from Theorem 4.1 and the fact that a central extension raises class by at most one.

Theorem 4.3 (Fifth- and sixth-depth stable class laws)**.** *For at-most-d-generated ordinary three-Engel algebras,*

$$H'' \subseteq Z_s(H) \quad \Longrightarrow \quad \operatorname{cl} H \le d+2 \quad \begin{cases} s=5, & d \ge 9, \\ s=6, & d \ge 10. \end{cases}$$

Both bounds are attained at every exact rank in these ranges. The fifth-depth start is least possible, since $\operatorname{cl} D_{7,5} = 10$ and $\operatorname{cl} D_{8,5} = 11$.

Full support gives each critical layer a determinant-twisted cubic representation. It makes singleton labels interchangeable and reduces the stable start to three finite profiles. A source-level suspension then propagates vanishing to all larger ranks, before applying Koszul multiplication. At the sixth depth an exterior-only reduction leaves just one profile after the repeated profile vanishes. Appendix D gives the general argument and exact starts. The least sixth-depth starting rank is not needed for the stated range.

Theorem 4.4 (The exceptional cubic layers)**.** *Let $T_d = (D_{d,5})_{d+3}$ with the natural action on marked generators and $U = k^d$. Then*

$$T_7 \simeq \det(U) \otimes \mathrm{Lie}^3(U), \qquad \dim T_7 = 112,$$
$$T_8 \simeq \det(U) \otimes \mathrm{Sym}^3(U), \qquad \dim T_8 = 120.$$

Here Lie^3 *is the cubic free ordinary Lie module and* Sym^3 *the ordinary symmetric cube. The rank-eight layer has a nonsplit submodule of dimension* 64 *and quotient of dimension* 56*, obtained by determinant twisting the Frobenius sequence* $0 \to U^{[1]} \otimes U \to \mathrm{Sym}^3 U \to \Lambda^3 U \to 0$.

The two ranks therefore have different class-raising parameter spaces, rather than merely different numbers of parameters. For $d \geq 9$ the entire fifth-depth critical layer is zero.

Corollary 4.5 (Delayed lifts over a complete free bottom)**.** *Over $E_d = D_{d,1}$, successive central epimorphisms of at-most-d- generated ordinary three-Engel algebras require at least four lifts to exceed class $d+2$ when $d \geq 6$, at least five when $d \geq 9$, and at least six when $d \geq 10$. Four suffice at ranks seven and eight; three suffice at rank five.*

Proof. After r ordinary central lifts the second derived ideal lies in Z_{r+1}. Apply the stable bounds. The boundary models and their marked fiber products with E_d give the attaining four-lift chains. Theorem 5.1 identifies the complete bottom. The ordinary current quotients in (38) provide the comparison objects, and Appendix D gives the boundary models and their common-bottom fiber products. □

For larger depths one also obtains

$$\mathrm{cl}\, H \leq \begin{cases} d+s-2, & d \geq 6,\ s \geq 4, \\ d+s-3, & d \geq 9,\ s \geq 5, \\ d+s-4, & d \geq 10,\ s \geq 6, \end{cases} \qquad H'' \subseteq Z_s(H),$$

by quotienting by the excess upper central depth. The classical centre-by-metabelian framework includes the balanced pairing and index-transfer identities of Kuz'min [2]; the corrected module treatments in [1, 3] fix the characteristic-two context. Here the balanced module controls successive central-action layers after the three-Engel relations are imposed.

5 Optimal centers in ordinary central realizations

Central realization asks for an algebra whose quotient by its entire center is a prescribed object. For the universal metabelian algebra H_d in (4), the smallest possible center has a sharp formula. The field in the ordinary statements below may be any characteristic-two field with more than two elements.

Theorem 5.1 (The least center in a central realization)**.** *For $d \geq 3$, put $h_d = (d-1)2^d + 1 + \binom{d}{2}$. Then*

$$\min_{E/Z(E) \simeq H_d} \dim_k Z(E) = \left\lceil \frac{d+1}{2} \right\rceil.$$

The minimum is over all finite-dimensional ordinary Lie realizations. It is attained by a d-generated three-Engel algebra of dimension $h_d + \lceil (d+1)/2 \rceil$, class $d+2$, and $Z(E) = E''$. For the complete object $E_d = D_{d,1}$,

$$Z(E_d) = E_d'', \qquad E_d/Z(E_d) \simeq H_d.$$

The lower bound is the classical injection $Z(H_d) \hookrightarrow \mathrm{Hom}_k(H_d/H_d', Z(E))$. The construction packs the two-generator central detectors and the top symmetric cocycle into exactly that many center coordinates. Its proof is in Appendix C. Optimizing the center is different from retaining the full kernel of a universal central cover.

6 Further questions

Let a_s be the least eventual starting rank of $\mathrm{cl}\, D_{d,s} = d+2$. The classical current quotients force $a_s \geq s+2$ for $s \geq 3$. We determine $a_5 = 9$ and prove $a_6 \leq 10$; determine a_6 and the growth of a_s. The finite-start criterion retains the degree-nine balanced source on a candidate diagonal. Uniform control of the outer commutator errors is the remaining issue, rather than a new finite rank table.

The present depth constraints give exact linear class laws in the stated ranges. Determining optimal class growth for unrestricted ordinary three-Engel algebras remains a separate problem. Likewise the minimum-center packing suggests questions for other prescribed ordinary quotients, where pair channels and top channels may interact differently.

A Proofs for the top cover and class boundary

A.1 The ordinary metabelian ceiling

We now work with ordinary Lie algebras over k and ordinary k-Lie generators. The algebra $H_d = U \oplus W_d$ is defined by (4), using exactly the module and triangle relations of Section 2.

Lemma A.1 (The classical metabelian ceiling)**.** *Every ordinary metabelian 3-Engel k-Lie algebra generated by at most $d \geq 2$ elements is a quotient of H_d. Its class is at most $\max(4, d+1)$, and H_d attains this bound at exact ordinary generator rank d.*

Proof. Let A be metabelian, put $W = A'$, and choose a linear section of A/W. On W, write $P_u = \mathrm{ad}_u|_W$ and let $\beta(u,v)$ be the bracket of the chosen lifts. The P_u commute. The 3-Engel identity gives $P_u^3 = 0$ and $P_u^2\beta(u,v) = 0$. The partial coefficients of $(P_u + \lambda P_v)^3 = 0$ give $P_u^2 P_v = P_u P_v^2 = 0$. The coefficient of λ in

$$P_{u+\lambda v}^2 \beta(u + \lambda v, w) = 0$$

then gives $P_u^2\beta(v,w) = 0$. These coefficient separations are valid over k: the polynomials have degree at most three, strictly less than $|k| = 4$. Since

$$W = \sum_v P_v W + \mathrm{span}_k\{\beta(v,w) : v, w \in A/W\},$$

we obtain $P_u^2 W = 0$ for every u. Conversely this square-zero action makes every third adjoint zero on A.

For any chosen d generators, the actions therefore factor through $R_d = k[t_i]/(t_i^2)$. Jacobi gives precisely the triangle relations (1), so the universal commutator-module presentation supplies a surjection $H_d \to A$. For H_d,

$$\gamma_2 H_d = W_d, \qquad \gamma_j H_d = \mathfrak{m}^{j-2} W_d \quad (j \geq 2).$$

Proposition 2.1 puts the highest coefficient degree at two for $d = 2, 3$, and at $d - 1$ for $d \geq 4$. It follows that $\operatorname{cl} H_d = \max(4, d+1)$. Its abelianization is U, so its ordinary generator rank is exactly d. This is the direct class consequence of the classical commutator module. □

A.2 The complete top central cover

The exterior-coinvariant description of free centre-by-metabelian central kernels is classical; see [1, (2.1) and Theorem 7.2]. Here we first impose the square-zero action relations of W_d and then compute the full top-weight CE quotient.

Central lifts, class, and ordinary generators. For an ordinary CE cocycle $\alpha : H \wedge H \to Z$, the central extension has bracket

$$[(u, a), (v, b)]_\alpha = ([u, v], \alpha(u, v)).$$

If H is 3-Engel, its extension is 3-Engel exactly when

$$\alpha([y, x, x], x) = 0 \qquad (x, y \in H), \tag{9}$$

because its third iterated bracket is $([y, x, x, x], \alpha([y, x, x], x))$. If H has class $c \geq 2$, projection of each lower-central term gives

$$\gamma_{c+1}(E_\alpha) = \operatorname{span}_k \alpha(\gamma_c H, H) \subseteq Z, \qquad \gamma_{c+2}(E_\alpha) = 0. \tag{10}$$

For $h \in \gamma_c H \subseteq Z(H)$, the CE identity implies $\alpha(h, H') = 0$; coboundaries also vanish on these top pairings. Thus a nonzero top pairing raises the class exactly once.

Let S be generated by arbitrary lifts of ordinary generators of H. Then $E_\alpha = S + Z$, and centrality gives $\gamma_j E_\alpha = \gamma_j S$ for $j \geq 2$. If $\dim Z = 1$ and the top pairing is nonzero, then $Z = \gamma_{c+1} E_\alpha \subseteq S$, so $E_\alpha = S$. Its ordinary generator rank therefore equals that of H.

Proof of Theorem 3.1. Write $W = W_d$, $R = R_d$, over k. The grading used here is Lie weight: $\deg x_i = 1$, $\deg v_{ij} = 2$, and $\deg t_i = 1$.

The top commutator space. Write $t_S = \prod_{i \in S} t_i$. The top graded part is W_{d+1}, where the subscript denotes the total Lie weight. It has a basis

$$T_i = t_{[d]\setminus\{j\}} v_{ij} \qquad (j \neq i), \tag{11}$$

and this expression does not depend on the choice of j. Indeed, the triangle relation on i, j, k, multiplied by $t_{[d]\setminus\{j,k\}}$, proves independence of j and k. If $s \notin \{i, j\}$, then

$$t_{[d]\setminus\{s\}} v_{ij} = 0. \tag{12}$$

To see this choose $k \notin \{i, j, s\}$, which is possible for $d \geq 4$, and multiply the triangle relation on i, j, k by $t_{[d]\setminus\{k,s\}}$. Its first two terms have a squared variable and vanish; the third is the displayed expression. These reductions show that the T_i span the top part.

Let $\theta : R \to k$ extract the coefficient of $t_1 \cdots t_d$. The map on top-degree presentations

$$\varphi(p v_{ij}) = \theta(p t_j) x_i + \theta(p t_i) x_j \qquad (\deg p = d - 1) \tag{13}$$

annihilates each triangle relation: each resulting term occurs twice. It takes T_i to x_i, proving independence. With $\Delta = x_1 \wedge \cdots \wedge x_d$, the canonical version is

$$W_{d+1} \simeq \det(U) \otimes U, \qquad T_i \longmapsto \Delta \otimes x_i.$$

Here naturality can be checked on generators of $\mathrm{GL}(U)$. Diagonal changes have weight $\det(B)B$ on the displayed T_i. Permutations permute their indices. For $x_1 \mapsto x_1 + x_2$, (11) with $j = 2$ gives $T_1 \mapsto T_1 + T_2$, while T_2 is fixed; for $i \notin \{1, 2\}$ the extra term vanishes by (12). Thus the determinant-twisted statement is independent of the basis.

Completeness of the CE quotient. Give a central coefficient space Z weight $d+2$. For an alternating CE cocycle $\alpha : H_d \wedge H_d \to Z$ homogeneous of total weight $d+2$, write

$$\eta(w,u) = \alpha(w,u), \qquad \beta(w,w') = \alpha(w,w').$$

The only possibly nonzero U–W terms are $\eta : W_{d+1} \otimes U \to Z$. The CE equations on x_i, w, w' give

$$\beta(t_i w, w') = \beta(w, t_i w'), \tag{14}$$

and those on x_i, x_j, w give

$$\beta(v_{ij}, w) = \eta(t_i w, x_j) + \eta(t_j w, x_i).$$

Thus every cocycle is uniquely determined by η, with

$$\beta(r v_{ij}, w) = \eta(t_i r w, x_j) + \eta(t_j r w, x_i), \tag{15}$$

where η is zero off W_{d+1}.

Put $B_{ij} = \eta(T_i, x_j)$. Symmetry of β forces $B_{ij} = B_{ji}$. For $j \neq l$, choose i, k so that i, j, k, l are distinct, and put $r = t_{[d]\setminus\{i,k\}}$. Equations (12)–(15) give

$$\beta(v_{ij}, r v_{kl}) = B_{lj}, \qquad \beta(r v_{kl}, v_{ij}) = B_{jl}.$$

This proves the required symmetry, with no condition on the diagonal.

Conversely, let $B : U \times U \to Z$ be any symmetric bilinear map after the displayed choice of Δ, and define η by its matrix B. Define β by (15). This descends from the free R-module to W: multiplication by each triangle relation gives six η terms, in three equal pairs, because the t_i commute. The square-zero ring relations are already respected. The expression can be written explicitly as

$$\begin{aligned}\beta(r v_{ij}, s v_{kl}) = {} & \theta(r s t_i t_l) B_{kj} + \theta(r s t_i t_k) B_{lj} \\ & + \theta(r s t_j t_l) B_{ki} + \theta(r s t_j t_k) B_{li}.\end{aligned} \tag{16}$$

This formula is symmetric under exchanging (ij, r) and (kl, s) because B is symmetric. It is alternating. More strongly,

$$\beta(pv, v) = 0 \qquad (p \in R,\ v \in W). \tag{17}$$

For one cyclic generator $v = r v_{ij}$, the four terms in (16) reduce to two squared-variable terms, which vanish, and $\theta(p r^2 t_i t_j)(B_{ij} + B_{ji}) = 0$. For a sum of cyclic generators the cross terms cancel in pairs, using symmetry and the balanced property (14). This proves (17), and in particular alternation.

Now set $\alpha|_{U\wedge U} = 0$, use η on $W \otimes U$, and use β on $W \wedge W$. All CE equations hold: the U, U, W equations are the definition of β; the U, W, W equations are balance; the W, W, W equations vanish because W is abelian; and the U, U, U equations vanish by the homogeneous weight. Thus every symmetric B extends, uniquely, to a homogeneous CE cocycle.

There are no coboundaries in this weight: H_d has no bracket of weight $d+2$, so $t([a,b])$ has zero component in this total weight. Consequently the homogeneous cohomology space is naturally

$$H^2_{\mathrm{CE}}(H_d, Z)_{d+2} \simeq \mathrm{Hom}_k(\det(U) \otimes \mathrm{Sym}^2 U, Z). \tag{18}$$

Automatic three-Engel admissibility. For $u, s \in U$ and $v, w \in W$, put $x = u + v$ and $y = s + w$. Write P_u for multiplication by the corresponding linear element of R, and $c(s, u) = [s, u]$. In the metabelian base,

$$[y, x, x] = P_u c(s, u) + P_u P_s v,$$

because $P_u^2 = 0$ on W. The central three-Engel obstruction is therefore

$$\begin{aligned}\alpha([y, x, x], x) = \eta(P_u c(s, u), u) + \eta(P_u P_s v, u) \\ + \beta(P_u c(s, u), v) + \beta(P_u P_s v, v).\end{aligned}$$

The first term has total weight four and vanishes because $d + 2 \geq 6$. Balance and (15) give

$$\beta(P_u c(s, u), v) = \beta(c(s, u), P_u v) = \eta(P_s P_u v, u) + \eta(P_u^2 v, s) = \eta(P_s P_u v, u).$$

It cancels the second term. The last term is zero by (17) with $p = P_u P_s$. Hence every cocycle in (18) satisfies $\alpha([y, x, x], x) = 0$ for every x, y.

Class and exact generator rank. When $\dim Z = 1$, a nonzero symmetric form has a nonzero top pairing. Equation (10) gives class $d + 2$, and the lifted generators generate the extension by the preceding central-lift argument. The ordinary rank is exactly d. The extension's second derived ideal is contained in Z. It is nonzero by Lemma A.1, since its class exceeds the metabelian ceiling $d + 1$. Hence it equals Z. □

A.3 The rank-two endpoint and the sharp boundary

Lemma A.2 (The ordinary two-generator endpoint)**.** *Over k, the free ordinary 3-Engel Lie algebra on two generators has dimension eight and class five. In particular $e(2) = 5$.*

Proof. Let F have basis a, b, c, u, v, w, z, t and nonzero unordered brackets

$$\begin{gathered}[a, b] = c, \quad [c, a] = u, \quad [c, b] = v, \quad [u, b] = [v, a] = w, \\ [w, a] = [c, u] = z, \quad [w, b] = [c, v] = t.\end{gathered} \tag{19}$$

All omitted brackets of distinct basis vectors vanish, and the bracket is alternating. In particular z, t are central. For Jacobi on distinct basis triples, the only triples with a nonzero individual nested term are $\{a, b, c\}$, $\{a, b, u\}$, and $\{a, b, v\}$. Their two nonzero terms are respectively $w + w$, $z + z$, and $t + t$. All three sums vanish. Triples with a repeated basis vector satisfy Jacobi automatically by alternation in characteristic two. This proves that the displayed table defines a Lie algebra.

For $x = Aa + Bb + Cc + Du + Ev + Fw + Gz + Tt$ and $R_x(y) = [y, x]$,

$$\begin{aligned}R_x(a) &= Bc + Cu + Ew + Fz, & R_x(b) &= Ac + Cv + Dw + Ft, \\ R_x(c) &= Au + Bv + Dz + Et, & R_x(u) &= Bw + Cz, \\ R_x(v) &= Aw + Ct, & R_x(w) &= Az + Bt, \\ R_x(z) &= R_x(t) = 0.\end{aligned}$$

Consequently

$$\begin{aligned}R_x^2(a) &= ABu + B^2 v + BCw + (BD + C^2 + AE)z, \\ R_x^2(b) &= A^2 u + ABv + ACw + (AE + C^2 + BD)t, \\ R_x^2(c) &= ACz + BCt, \\ R_x^2(u) &= ABz + B^2 t, \qquad R_x^2(v) = A^2 z + ABt, \\ R_x^2(w) &= R_x^2(z) = R_x^2(t) = 0.\end{aligned}$$

Substitution of the first table into the second gives $R_x^3 = 0$: the noncentral terms cancel in pairs, and z, t are central. Thus F is ordinary three-Engel. Its lower central dimensions are

$$(8, 6, 5, 3, 2, 0),$$

because the successive spans are F, $\langle c, u, v, w, z, t\rangle$, $\langle u, v, w, z, t\rangle$, $\langle w, z, t\rangle$, $\langle z, t\rangle$, and zero. It has exact ordinary rank two and $F'' = \langle z, t\rangle \neq 0$.

Every pair of elements has this table. Let L be any ordinary three-Engel Lie algebra over k, and fix $a, b \in L$. Define

$$c = [a, b], \quad u = [c, a], \quad v = [c, b], \quad w = [u, b] = [v, a], \quad z = [w, a], \quad t = [w, b].$$

The equality defining w is Jacobi. Engel gives

$$[u, a] = [v, b] = 0, \qquad [z, a] = [t, b] = 0.$$

For example $[z, a] = [v, a, a, a] = 0$, and the other last equality is $[u, b, b, b] = 0$. Jacobi also gives

$$[c, u] = z, \qquad [c, v] = t.$$

Set

$$p = [z, b], \qquad q = [t, a], \qquad r = [c, w], \qquad s = [u, v].$$

Jacobi on $\{c, u, b\}$, $\{c, v, a\}$, and $\{a, b, w\}$ gives

$$p + r + s = 0, \qquad q + r + s = 0, \qquad r + q + p = 0.$$

Since the characteristic is two, these imply

$$p = q, \qquad r = 0, \qquad s = p.$$

It remains to kill p. Apply the three-Engel identity to $y = a$ and $x = b + \lambda c$:

$$\begin{aligned} 0 &= [a, b + \lambda c, b + \lambda c, b + \lambda c] \\ &= [v + \lambda w + \lambda^2 z, b + \lambda c] \\ &= \lambda^2(r + p) + \lambda^3[z, c]. \end{aligned}$$

For $\lambda \neq 0$, division by λ^2 yields $r + p + \lambda[z, c] = 0$. Two distinct nonzero values of λ exist because $|k| > 2$. Subtraction first gives $[z, c] = 0$, then $r + p = 0$. As $r = 0$, it follows that $p = q = r = s = 0$.

Thus z, t commute with a, b, and are central in the subalgebra generated by a, b: the centralizer of either element is a Lie subalgebra containing the generators. Finally

$$[u, w] = [u, [v, a]] = [[u, v], a] + [v, [u, a]] = 0,$$

and similarly $[v, w] = [v, [u, b]] = [[v, u], b] + [u, [v, b]] = 0$. All unordered brackets of a, b, c, u, v, w, z, t are now exactly those in (19). Their span is a Lie subalgebra containing a, b, and hence equals $\langle a, b\rangle_L$. The linear map from F sending its two generators to this arbitrary pair is therefore a Lie homomorphism. It is unique since a, b generate F. This proves the claimed free universal property and the exact class maximum.

□

Proof of Theorem 3.2. Let L have ordinary generator rank at most $d \geq 2$, satisfy the 3-Engel identity, and have $L'' \subseteq Z(L)$. Then $L/Z(L)$ is metabelian, 3-Engel, and generated by at most d elements. Lemma A.1 gives

$$\operatorname{cl} L \leq \operatorname{cl}(L/Z(L)) + 1 \leq \max(4, d+1) + 1 = \max(5, d+2).$$

If the quotient is zero, the class is at most one. Otherwise the first inequality follows by lifting its lower-central series: the next term lies in $Z(L)$, and one more bracket kills it.

For $d \geq 4$, any nonzero one-dimensional top cover from Theorem 3.1 has exact ordinary rank d, class $d+2$, and central second derived ideal. For $d = 2$, the eight-dimensional algebra in Lemma A.2 has class five and central second derived ideal. Its direct sum with a central one-dimensional abelian algebra attains class five at exact rank three. These models prove sharpness at both exact rank and rank at most d. □

B Central-depth bounds and their exact starting systems

This appendix proves Theorem 4.1 and Proposition 4.2. Work over a characteristic-two field k with $|k| > 2$ and use left-normed brackets. Set $K = H''$, $M_0 = K$ and $M_{i+1} = [M_i, H]$. Write $F_d = \mathcal{F}_d$ for the relatively free ordinary three-Engel algebra on d degree-one generators, as in Section 4. For $U = k^d$, use

$$R = \Lambda(U), \qquad P_d = \langle c_{ij} = c_{ji},\ c_{ii} = 0 \mid t_i c_{jk} + t_j c_{ik} + t_k c_{ij} = 0 \rangle_R, \qquad B_d = \operatorname{Sym}^2_R(P_d).$$

The grading is $\deg t_i = 1$, $\deg c_{ij} = 2$. Write $P = P_d$ and $B = B_d$ when the rank is fixed. The tensor product defining B_d is over R; it is distinct from a tensor product over k carrying a diagonal derivation action. The arguments use the classical commutator and Koszul constructions [2, 1, 3].

B.1 The metabelian presentation and its top

In a metabelian three-Engel algebra, the actions on the derived ideal commute. The coefficient of two copies of x and one copy of z in $[y, x+\lambda z, x+\lambda z, x+\lambda z] = 0$ gives

$$[y, z, x, x] + [y, x, z, x] + [y, x, x, z] = 0.$$

The last two terms agree modulo H'', so $t_x^2(H'/H'') = 0$. Jacobi supplies the displayed triangle relation. Conversely $U \oplus P$ with $[u, v] = c(u, v)$ and $[w, u] = t_u w$ is metabelian and three-Engel: after the first bracket, the next two actions give $t_x^2 = 0$. Thus this is the free metabelian object in question.

For $d \geq 4$, $P_n = 0$ for $n > d+1$. The only possible coefficient in degree $d+2$ is $t_1 \cdots t_d$, and it kills every c_{ij}: multiply the triangle on i, j, k by $t_i t_j$ for a distinct k, then multiply by the remaining variables. Lower-support cases are covered by the same argument and the degree bound for the active ring. The degree-$d+1$ elements reduce to

$$T_i = t_{[d]\setminus\{j\}} c_{ij} \quad (j \neq i).$$

Triangles identify the choices of j, and

$$\iota : P \to R \otimes U, \qquad c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i$$

takes T_i to $(t_1 \cdots t_d) \otimes x_i$. Therefore $P_{d+1} \cong \det(U) \otimes U$ and the metabelian class ceiling is $d+1$ in these ranks. This argument also shows the finite presentation being used. The argument proves injectivity on the displayed graded top and does not require global injectivity.

B.2 The top symmetric square of the Koszul presentation

The following module statement has a direct, rank-uniform proof:

$$(B_d)_{d+2} \cong \det U \otimes \operatorname{Sym}^2 U \quad (d \geq 5), \qquad (B_d)_n = 0 \quad (n > d+2,\ d \geq 4). \tag{20}$$

For the second assertion, degree $d+3$ suffices since B_d is generated in degree four over R. Proceed by induction on the active support. Supports of size at most three have no degree above six: in rank three, a coefficient using all three variables kills every c_{ij} by in the presentation of P_d; ranks one and two are immediate. For active support $r \geq 4$, every exponent in a surviving symbol is at most three. In degree $r+3$ only the patterns $(3,2,1,\dots,1)$ and $(2,2,2,1,\dots,1)$ remain.

For the first pattern, with repeated indices i,j, the symbols are

$$D = c_{ij} \odot c_{ij}\, t_{[r]\setminus\{j\}}, \qquad X_k = c_{ij} \odot c_{ik}\, t_{[r]\setminus\{k\}} \quad (k \notin \{i,j\}).$$

Triangle (i,j,k) paired with c_{ij} gives $X_k = D$. Triangle (i,j,k) paired with c_{il}, with four distinct indices, gives $X_l = 0$: its other two terms have squared coefficients. Thus all these symbols vanish.

For the second pattern, with repeated indices i,j,k, the symbols are

$$A_i = c_{ij} \odot c_{ik}\, t_{[r]\setminus\{i\}} \quad \text{and cyclic variants},$$
$$B_{i,a} = c_{ia} \odot c_{jk}\, t_{[r]\setminus\{a\}} \quad (a \notin \{i,j,k\}),$$

with cyclic variants. Triangle (i,j,k) paired with c_{ia} gives $B_{i,a} = 0$; triangle (i,j,a) paired with c_{ik} gives $B_{j,a} = A_i$. A spare fourth index exists for $r \geq 4$, so all symbols vanish. This completes the induction and the upper-degree bound.

For degree $d+2$ and $d \geq 5$, that bound also shows that every component with fewer than d active variables is zero. The full-support patterns are precisely $(3,1,\dots,1)$ and $(2,2,1,\dots,1)$.

For the first pattern fix the tripled index i. Its symbols are

$$X_{jk} = c_{ij} \odot c_{ik}\, t_{[d]\setminus\{j,k\}} \quad (j,k \neq i,\ j \neq k).$$

Triangle (i,j,k) paired with c_{ih}, for distinct $j,k,h \neq i$, gives $X_{kh} = X_{jh}$. All symbols are therefore equal.

For the second pattern fix the doubled indices i,j and put $O = [d] \setminus \{i,j\}$. The symbols are

$$D = c_{ij} \odot c_{ij}\, t_O,$$
$$A_a = c_{ij} \odot c_{ia}\, t_{[d]\setminus\{i,a\}}, \qquad B_a = c_{ij} \odot c_{ja}\, t_{[d]\setminus\{j,a\}},$$
$$C_{ab} = c_{ia} \odot c_{jb}\, t_{[d]\setminus\{a,b\}}, \qquad E_{ab} = c_{ij} \odot c_{ab}\, t_{[d]\setminus\{a,b\}},$$

where $a,b \in O$, $a \neq b$, and $E_{ab} = E_{ba}$. Triangles give

$$D = A_a + B_a, \qquad C_{ab} = A_a = B_b, \qquad E_{ab} = A_a + A_b.$$

With three distinct indices $a,b,c \in O$, triangle (i,a,b) paired with c_{jc} gives $C_{bc} = C_{ac}$, hence $A_b = A_a$. Since $|O| \geq 3$, all A_a, B_a, C_{ab} have one common value and $D = E_{ab} = 0$. Each full-support component has dimension at most one.

To prove nonvanishing and naturality, use the Koszul map

$$\iota : P_d \to R \otimes U, \qquad c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i.$$

Multiplication defines

$$\mu : \operatorname{Sym}^2_R(P_d) \to R \otimes \operatorname{Sym}^2 U.$$

In degree $d+2$ its target is $R_d \otimes \operatorname{Sym}^2 U = \det U \otimes \operatorname{Sym}^2 U$. The X_{jk} map to $(t_1 \cdots t_d) \otimes x_i^2$ and the A_a map to $(t_1 \cdots t_d) \otimes x_i x_j$. This proves (20), including the natural module identification. In characteristic two the symmetric square here is not silently identified with a space of divided powers.

B.3 Peeling an operator into the first outer input

Let $1 \le r \le 3$, and assume $M_{r+1} = 0$. Jacobi gives $[M_i, H'] \subseteq M_{i+2}$, so

$$\Theta_r(w, v; a_1, \dots, a_r) = [\widetilde{w}, \widetilde{v}, a_1, \dots, a_r]$$

is well defined on $(H'/K)^2 \times U^r$. It is balanced in the two derived legs. The partial Engel identity with inputs a, a, v, followed by $a_2, \dots, a_r$, gives

$$\Theta_r(t_a w, v; a, a_2, \dots, a_r) = 0.$$

Indeed its other terms belong to M_{r+1}: the term involving $[w, a, a] \in K$ first brackets with $v \in H'$, landing in M_2, and then has $r - 1$ outer actions. Polarization exchanges an operator with the first outer input.

Consequently, for $N \ge r + 5$, there is a natural epimorphism

$$B_{N-r+1} \otimes U^{\otimes(r-1)} \twoheadrightarrow (M_r)_N. \tag{21}$$

For a square-free coefficient t_S, peel one factor $a \in S$ and send a symbol to

$$t_S(c_{ij} \odot c_{kl}) \otimes a_2 \otimes \cdots \otimes a_r \longmapsto \Theta_r(t_{S\setminus\{a\}} c_{ij}, c_{kl}; a, a_2, \dots, a_r).$$

The exchange makes the choice independent, kills squared coefficients, and respects each triangle relation by peeling from its common coefficient, of degree $N - r - 4 \ge 1$. This proves a quotient map; it does not put an unproved residual R-action on an exterior quotient.

The second-layer ceiling. In the free graded algebra with $M_2 = 0$, the quotient by M_1 is centre-by-metabelian. Its quotient by the centre is metabelian and has class at most $d + 1$, hence it has class at most $d + 2$. Every homogeneous part in degree $N > d + 2$ is therefore in M_1. But (21) for $r = 1$ is an epimorphism $B_N \twoheadrightarrow (M_1)_N$, and (20) kills B_N. Thus the free depth-two object already has class at most $d + 2$ for every $d \ge 4$. This elementary proof needs neither the full top central cohomology classification nor an external unrestricted Engel class bound.

B.4 An explicit collecting identity for the third layer

Work in characteristic two over a field with more than three elements. Brackets are left normed. Set $v = [p, a]$, $b = [p, q]$, and define

$$E_{21}(y; x, z) = [y, x, x, z] + [y, x, z, x] + [y, z, x, x],$$

$$E_{111}(y; x, z, t) = \sum_{\sigma \in S_3} [y, x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}], \qquad (x_1, x_2, x_3) = (x, z, t).$$

Both vanish in an ordinary three-Engel algebra: they are the complete cubic coefficients, whose extraction is valid because $3 < |k|$. The permutations in the second formula are indexed permutations, even when some substituted values happen to coincide.

Write

$$D(x_1, x_2, x_3, x_4; x_5, x_6, x_7) = [[[x_1, x_2], [x_3, x_4]], x_5, x_6, x_7].$$

Let $D_1, \dots, D_{13}$ be the following displayed values or brackets of values. Each is zero if $[H'', H, H, H] = 0$.

1	$D(p, a, [p, q], q; h, r, r)$
2	$D(p, a, [p, q], q; r, r, h)$
3	$[D(p, q, p, a; r, r, q), h]$
4	$[D(p, a, h, r; p, q, q), r]$
5	$[D(p, a, h, r; q, q, p), r]$
6	$D([p, q], q, h, r; r, p, a)$
7	$D([p, q], q, h, r; r, a, p)$
8	$D(h, r, [p, a], r; p, q, q)$
9	$D(h, r, [p, a], r; q, q, p)$
10	$D(q, r, [p, q], r; p, a, h)$
11	$D(q, r, [p, q], r; a, p, h)$
12	$D(p, a, [p, q], r; q, r, h)$
13	$D(p, q, q, r; [p, a], r, h)$

The following equality holds already in the free ordinary Lie algebra in characteristic two:

$$[[[p, q], q, r], [p, a], h, r] = [E_{21}(v; r, b), q, h] + [E_{111}(r; q, v, [b, r]), h] + \sum_{i=1}^{13} D_i. \tag{22}$$

It is an explicit collecting identity, using only the Lie relations. It can be checked by expanding commutators; `verification/central_depth/verify_sparse_seed.py` does this independently, without a quotient-space or rank calculation. Under the three-Engel and depth-three hypotheses every summand on the right vanishes, proving the seed.

B.5 The third-layer source and its vanishing

In the free depth-three object put $T_d = (M_2(F_d)/M_3(F_d))_{d+3}$. The preceding module calculation and peeling construction give a natural epimorphism

$$\Psi_d : (\det U \otimes \operatorname{Sym}^2 U) \otimes U \twoheadrightarrow T_d, \qquad U = k^d, \quad d \geq 5. \tag{23}$$

Its ingredients are the metabelian presentation over $R = \Lambda(U)$, the central pairing

$$\Theta(w, v; a, b) = [\widetilde{w}, \widetilde{v}, a, b],$$

balance $\Theta(t_a w, v; h, z) = \Theta(w, t_a v; h, z)$, and the exchange

$$\Theta(t_a w, v; h, z) = \Theta(t_h w, v; a, z).$$

It identifies the degree-$d+2$ part of $\operatorname{Sym}^2_R(P_d)$ with $\det U \otimes \operatorname{Sym}^2 U$ via the Koszul multiplication

$$c_{ij} \longmapsto t_i \otimes x_j + t_j \otimes x_i.$$

Only the source in (23) is a symmetric tensor module: T_d is a quotient, whose extra relations must still be proved.

Fix distinct coordinate indices p, q, a, h, and put $S = [d] \setminus \{p, a\}$, with $h \in S \setminus \{q\}$. For any coordinate generator z, the word

$$P^{(d)}_{pq,z} = \Theta(t_{S \setminus \{h\}} c_{pq}, c_{pa}; h, z) \tag{24}$$

is the image of $(t_1 \cdots t_d) \otimes x_p x_q \otimes z$ under (23). This follows directly by multiplying the two Koszul images.

The rank-five case. Equation (22) gives the universal identity

$$[[[p,q],q,r],[p,a],h,r]=0. \tag{25}$$

It holds in every ordinary three-Engel algebra with $M_3=0$. The five formal variables are p,q,a,h,r. The multidegree is $(2,2,1,1,2)$. The explicit identity above and its free-associative expansion certify all its Lie inputs.

For $d=5$, (25) is exactly (24) with $z=r$ distinct from p,q. It kills the image of one tensor $x_px_q\otimes x_r$ with three distinct indices. Permutation matrices kill all such coordinate tensors. A transvection $x_r\mapsto x_r+x_p$, fixing x_p,x_q, then kills $x_px_q\otimes x_p$; similarly it kills the tensor with last input x_q. Hence every off-diagonal tensor $x_px_q\otimes z$ is killed.

Finally, for any q,z, choose $p\notin\{q,z\}$. The transvection $x_p\mapsto x_p+x_q$ fixes z and has determinant one. Applied to the already killed tensor $x_px_q\otimes z$, it kills $x_q^2\otimes z$. Therefore the entire source of (23) is killed, and $T_5=0$.

Propagation to arbitrary rank. Assume $T_d=0$ for some $d\geq 5$. Add a new generator y. For old p,q,z, choose a,h as in (24). Since (24) is zero in the relatively free depth-three algebra on the old generators, it remains zero after every substitution

$$x_p\longmapsto x_p+\lambda[x_p,y],\qquad \lambda\in k.$$

There are two occurrences of x_p in (24), or three if $z=x_p$. Thus this is a vector-valued polynomial in λ of degree at most three. A polynomial of degree at most three vanishing on all four elements of $\mathbb{F}_4$ has all coefficients zero. In particular its coefficient of λ vanishes. The field-size check is necessary here; an unqualified extraction from a degree-four polynomial on $\mathbb{F}_4$ would be invalid.

The variable p is absent from the operator coefficient $t_{S\setminus\{h\}}$ and from the first outer input h. Modulo $K=H''$, the two head variations are

$$\delta c_{pq}=t_yc_{pq}+t_pc_{qy},\qquad \delta c_{pa}=t_yc_{pa}+t_pc_{ay}. \tag{26}$$

The two t_y terms cancel by balance. If $z=x_p$, its variation lies in H' and gives zero in the last outer slot of Θ. Changes of head lifts by K also give zero. The surviving coefficient is

$$\Theta(t_pt_{S\setminus\{h\}}c_{qy},c_{pa};h,z)+\Theta(t_pt_{S\setminus\{h\}}c_{pq},c_{ay};h,z). \tag{27}$$

Under the rank-$(d+1)$ symmetric-module identification, (27) is the image of

$$t_{[d]\setminus\{a\}}\big(c_{qy}\odot c_{pa}+c_{pq}\odot c_{ay}\big)\otimes z.$$

In the notation of the module calculation above these are respectively $C_{a,y}$ and $E_{a,y}$. Their images under Koszul multiplication are x_px_q and zero. Consequently (27) is exactly

$$\Psi_{d+1}((t_1\cdots t_dt_y)\otimes x_px_q\otimes z). \tag{28}$$

It vanishes by coefficient extraction.

For any off-diagonal tensor in rank $d+1$, one can choose the new generator y outside $\{p,q,z\}$ and apply this argument to the remaining d generators. Hence every off-diagonal tensor is killed. The determinant-one transvection used in Step 2 then kills every diagonal tensor. Surjectivity of (23) proves $T_{d+1}=0$.

This is an induction on arbitrary rank, not an extrapolation from ranks five and six. Equations (26)–(28), including the zero E term, are proved in the rank-uniform module presentation.

The class bound. The previous second-central-layer theorem gives $\operatorname{cl} D_{d,2} = d+2$ for $d \geq 4$. Since $D_{d,3} \to D_{d,2}$ is central, $\operatorname{cl} D_{d,3} \leq d+3$. Any degree-$d+3$ component is contained in $M_2/M_3 = T_d$. The vanishing just proved therefore improves the upper bound to $d+2$. The previously constructed centre-by-metabelian top covers have exact generator rank d and class $d+2$, and already satisfy $H'' \subseteq Z(H)$. They give sharpness. Every at-most-d-generated algebra in the stated variety is a quotient of $D_{d,3}$, completing the third-layer case of Theorem 4.1.

B.6 The next-to-top balanced symmetric square

For $U = k^d$, write $R = \Lambda U$ and

$$P_d = \langle c_{ij} = c_{ji},\ c_{ii} = 0 \mid t_i c_{jk} + t_j c_{ik} + t_k c_{ij} = 0 \rangle_R, \qquad B_d = \operatorname{Sym}^2_R P_d.$$

Use $\deg t_i = 1$, $\deg c_{ij} = 2$. The Koszul multiplication $\mu : B_d \to R \otimes \operatorname{Sym}^2 U$ is induced by $c_{ij} \mapsto t_i x_j + t_j x_i$. For $d \geq 6$ it gives a natural identification

$$(B_d)_{d+1} \simeq \mathcal{K}_d := \ker\left(\Lambda^{d-1} U \otimes \operatorname{Sym}^2 U \xrightarrow{\partial} \det U \otimes U\right), \tag{29}$$

where

$$\partial(r \otimes xy) = (r \wedge x) \otimes y + (r \wedge y) \otimes x.$$

In particular

$$\dim \mathcal{K}_d = \frac{d(d-1)(d+2)}{2}.$$

This is a specific truncated-module computation using classical Koszul and symmetric-power constructions.

Proof of (29)**.** The bound (20) $(B_r)_n = 0$ for $n > r+2$, together with the rank-at-most-three calculation, shows that degree $d+1$ has support of size $d-1$ or d. On a support of size $d-1 \geq 5$ the previous top theorem identifies the component with $\det(k^{d-1}) \otimes \operatorname{Sym}^2(k^{d-1})$ and μ is injective.

For full support, the multidegree has one doubled index i. Put $O = [d] \setminus \{i\}$. Its generators are

$$X_{ab} = c_{ia} c_{ib}\, t_{[d]\setminus\{i,a,b\}}, \qquad Y_{a;bc} = c_{ia} c_{bc}\, t_{[d]\setminus\{a,b,c\}},$$

with distinct indices in O. Triangles give $Y_{c;ab} = X_{ac} + X_{bc}$ and, for four distinct indices in O,

$$X_{ac} + X_{ad} + X_{bc} + X_{bd} = 0. \tag{30}$$

The edge space on O modulo the rectangle relations (30) has dimension at most $|O|$. One way to see this is to dualize: edge weights satisfying every rectangle relation have the form $f_{ab} = u_a + u_b + c$, a space of dimension $|O|$. The images of X_{ab} are

$$Q_i + P_{i,a} + P_{i,b}, \qquad Q_i = t_{\widehat{i}} \otimes x_i^2, \quad P_{i,a} = t_{\widehat{a}} \otimes x_i x_a.$$

They have rank $|O|$: a triangle yields Q_i, and differences $P_{i,a} + P_{i,b}$ have rank $|O| - 1$. Thus μ is injective also on full support. Here $t_{\widehat{j}}$ is the product of all variables except t_j.

The kernel of ∂ has the following complete coordinate description:

- $t_{\widehat{j}} \otimes x_a x_b$ with $a, b \neq j$;
- for each doubled i, Q_i and $P_{i,a} + P_{i,b}$ with $a, b \neq i$.

Each target monomial is either one of the first items, Q_i, or $P_{i,a}$; on the latter ∂ is $(t_1 \cdots t_d) \otimes x_i$. This proves both the completeness of the displayed kernel and (29).

B.7 The depth-four source and nonlinear suspension

If $M_4 = 0$, set

$$\Theta_3(w, v; a, b, c) = [\widetilde{w}, \widetilde{v}, a, b, c].$$

As in the third-layer proof, $[M_j, H'] \subseteq M_{j+2}$ shows that the derived inputs factor through H'/H'' and the outer inputs through H/H'. Balance and the exchange of an operator with the first outer input follow from Jacobi and the partial Engel identity, now with two outer brackets appended. Peeling therefore gives, for $N \geq 8$,

$$(B_d)_{N-2} \otimes U \otimes U \twoheadrightarrow (M_3)_N.$$

At $N = d + 3$, (29) gives the natural epimorphism

$$\Phi_d : \mathcal{K}_d \otimes U^{\otimes 2} \twoheadrightarrow T_d := (M_3/M_4)_{d+3}. \tag{31}$$

Assume $T_d = 0$ and add a new generator y. Substitute $x_i \mapsto x_i + \lambda[x_i, y]$ in any old source word. On H'/H'', the coefficient operators do not change, because $[x_i, y] \in H'$. The outer inputs do not change in H/H'. Only the two commutator heads change. Consequently the resulting expression in the central target has degree at most two in λ, regardless of the number of occurrences of i in the original source word. Coefficient extraction over $\mathbb{F}_4$ is therefore valid. This detail prevents a false extraction from a raw degree-four polynomial.

The coefficient of λ induces on Koszul images the map

$$\sigma_{i,y}(z) = (t_y x_i + t_i x_y)\frac{\partial z}{\partial x_i}, \tag{32}$$

with the coefficient $r \in R$ held fixed. For example,

$$\delta_i c_{ij} = t_y c_{ij} + t_i c_{jy}, \qquad \mu(\delta_i c_{ij}) = t_j(t_y x_i + t_i x_y).$$

Triangle relations are respected: their two residual terms cancel by the triangle on (j, k, y). Thus (32) defines a map on the balanced symmetric module, not just on a list of representatives. For u, v in the old generator space,

$$\Phi_{d+1}(\sigma_{i,y}(z) \otimes u \otimes v) = 0. \tag{33}$$

B.8 The suspension images span the next source

Let $n = d + 1 \geq 7$. Let Y be the $\mathrm{GL}_n(k)$-stable span of the tensors in (33), allowing every omitted coordinate y. It lies in $\ker \Phi_n$. We show $Y = \mathcal{K}_n \otimes U^{\otimes 2}$. Fix coordinate outer inputs u, v throughout each construction.

First consider $t_{\widehat{j}} \otimes x_a x_b$, where $a, b \neq j$ and $a \neq b$. Choose $y \notin \{j, a, b, u, v\}$. In the old space use $z = t_{\widehat{j,y}} \otimes x_a x_b$ and apply $\sigma_{a,y}$. Since its coefficient already contains t_a, (32) gives exactly $t_{\widehat{j}} \otimes x_a x_b$. Thus all these tensors belong to Y. Their square counterparts follow by a shear $x_a \mapsto x_a + x_b$, choosing $a \notin \{j, b, u, v\}$ so that the coefficient and the outer inputs are fixed. Hence all $t_{\widehat{j}} \otimes x_b^2$, $b \neq j$, are in Y as well.

For $P_{i,a} + P_{i,b}$ choose $y \notin \{i, a, b, u, v\}$. In the old space the element $Q_i + P_{i,a} + P_{i,b}$ belongs to $\mathcal{K}_d$. Equation (32) with respect to x_i kills Q_i and sends the other two terms to the desired $P_{i,a} + P_{i,b}$ in rank n. Thus all these differences are in Y.

Finally fix the desired Q_j, and choose $i \notin \{j, u, v\}$. The already available missing-support tensor $t_{\widehat{j}} \otimes x_i^2 \otimes u \otimes v$ transforms under $x_i \mapsto x_i + x_j$ into the sum of itself, $t_{\widehat{i}} \otimes x_j^2 \otimes u \otimes v$, and $(Q_i + Q_j) \otimes u \otimes v$. The first two are known to lie in Y, hence so does the last. Scaling x_i by $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$ gives $(\omega^2 Q_i + \omega Q_j) \otimes u \otimes v$. The outer inputs are fixed by the choice of i. Since $\omega^2 + \omega = 1$, these two vectors separate Q_i and Q_j. This proves the spanning assertion without assuming semisimplicity. All choices are possible for $n \geq 7$.

Together with (31)–(33), this proves

$$T_d = 0 \implies T_{d+1} = 0 \qquad (d \geq 6). \tag{34}$$

B.9 The finite base and the rank-five boundary

The previous third-layer result gives $\operatorname{cl} D_{d,3} = d+2$ for $d \geq 5$. Since $D_{d,4} \to D_{d,3}$ is central, $\operatorname{cl} D_{d,4} \leq d+3$. In rank six the only full-support degree-nine patterns are

$$(4,1,1,1,1,1), \quad (3,2,1,1,1,1), \quad (2,2,2,1,1,1).$$

Each corresponding capped graded universal extension calculation has zero top kernel. The final cochain matrices have respectively 458, 1299, and 2218 columns and full rank. The exact input tables and independent replay are supplied in `verification/central_depth/`. These are complete homogeneous cubic coefficients over $\mathbb{F}_4$, not samples at binary points.

Components supported on at most five generators vanish because $D_{5,4}$ has class at most eight. Therefore $T_6 = 0$. Induction (34) proves $T_d = 0$ for every $d \geq 6$, and centrality then improves the upper class bound from $d+3$ to $d+2$. The attaining models of Theorem 3.1 give equality.

For rank five, the supplied 178-dimensional capped model has

$$(\dim M_0, \ldots, \dim M_4) = (109, 60, 39, 11, 0)$$

and class eight. Its word $[[[p,q],p],[p,a],h,p,r]$ is nonzero. Its quotients by M_1, M_2, M_3 all have class seven and exact generator rank five. They give a concrete $7 \to 7 \to 7 \to 8$ chain of three central lifts. The kernels need not be one-dimensional. The same lower example and the central upper bound prove $\operatorname{cl} D_{5,4} = 8$ for the complete relatively free object.

Completeness of the finite starting step. At each weight, the replay enumerates all alternating cochains allowed by the multidegree cap, imposes Jacobi, every coefficient of the cubic Engel identity, and $M_4 = 0$, and computes their common kernel. It also checks that the actual added bracket coordinates are independent and span this entire kernel. Induction on weight therefore identifies the capped universal component, rather than merely verifying a model. The three degree-nine kernels vanish. The independent implementation uses the opposite pivot order and does not import the constructor. Complete cubic coefficients are required over $\mathbb{F}_4$; testing only binary-coordinate inputs would not certify the same variety.

The rank-four boundary. The supplied 40-, 41-, and 42-dimensional bracket tables form central quotients of exact rank four and classes $6, 6, 7$. Direct exact checks impose Jacobi and the complete cubic Engel coefficients. Their second derived ideals have central depths at most one, two, and three, respectively. The second-layer bound and the one-step central upper bound prove the rank-four entries of Proposition 4.2. These lower witnesses and the rank-five model establish every stated boundary value.

C Optimal central realizations

Write k for a characteristic-two field with $|k| > 2$ in the ordinary part of the argument.

C.1 The center and the lower bound

Use the presentation $H_d = U \oplus P_d$ of Appendix B. Its center, for $d \geq 3$, is

$$Z(H_d) = \mathcal{S}_d \oplus \mathcal{T}_d, \qquad \mathcal{S}_d = \operatorname{span}\{t_i t_j c_{ij} : i < j\}, \qquad \mathcal{T}_d = \operatorname{span}\{T_i : 1 \leq i \leq d\}.$$

Here $T_i = t_{[d]\setminus\{j\}} c_{ij}$ is independent of $j \neq i$. The summands have dimensions $\binom{d}{2}$ and d. For a direct socle calculation, use the Koszul map $c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i$. On a multilinear support

I, the triangle presentation is the edge space modulo triangle cycles, of dimension $|I| - 1$. Its image has coordinates $t_{I\setminus\{i\}} \otimes x_i$ with coefficient sum zero. Multiplication by t_i detects the ith coordinate, so no nonzero multilinear vector is central. With one doubled index i, triangles identify all partners, and the image is the line $kt_I \otimes x_i$. A missing variable acts nontrivially; only $I = [d]$ is central. With two doubled indices i, j, the symbol is $t_I c_{ij}$. A third index in I kills it by the triangle on i, j and that index. The surviving $t_i t_j c_{ij}$ is central, and projection onto H_2 detects it. These projections prove independence of all such vectors; the Koszul images prove independence of the T_i. This proves the center formula, including rank three where both summands have weight four. The same presentation gives

$$\dim H_d = (d-1)2^d + 1 + \binom{d}{2} = h_d, \qquad \operatorname{cl} H_d = d + 1.$$

Suppose $K = Z(E)$ and $E/K = H_d$. Commuting lifts define

$$\Psi : Z(H_d) \longrightarrow \operatorname{Hom}_k(H_d/H_d', K), \qquad \Psi(z)(x) = [\widetilde{z}, \widetilde{x}].$$

Jacobi shows that a lift of z commutes with lifts of H_d', so the map is well defined. If its value is zero, the lift is central and $z = 0$. Thus $d \dim K \geq d(d+1)/2$, proving the lower bound.

C.2 Pair detectors and the top detector

The rank-two algebra on a, b, c, u, v, w, z with nonzero brackets

$$[a, b] = c, \quad [a, c] = u, \quad [b, c] = v, \quad [a, v] = [b, u] = w, \quad [a, w] = [c, u] = z$$

is three-Engel, with center and second derived ideal kz. Its quotient is the six-dimensional H_2. The corresponding scalar cocycle has values $\zeta(a, w) = \zeta(c, u) = 1$. Its complete cubic coefficients vanish; equivalently $\zeta([y, x, x], x) = 0$ for every $x, y \in H_2$. This is also the rank-two endpoint in Appendix A.

For an ordered pair $i \to j$, pull this cocycle back along $x_i \mapsto a, x_j \mapsto b$, with all other generators zero. Call it ζ_{ij}^i. It detects exactly

$$\zeta_{ij}^i(t_i t_j c_{ij}, x_i) = 1;$$

all other center–generator pairings are zero. The top vectors T_l have full support, so their two-generator projections vanish.

Let $\theta : R \to k$ extract $t_1 \cdots t_d$. The identity symmetric form gives the top cocycle

$$\begin{aligned} \alpha_I(x_i, x_j) &= 0, \\ \alpha_I(rc_{ij}, x_l) &= \theta(rt_j)\delta_{il} + \theta(rt_i)\delta_{jl}, \\ \alpha_I(rc_{ij}, sc_{kl}) &= \theta(rst_it_l)\delta_{kj} + \theta(rst_it_k)\delta_{lj} \\ &\quad + \theta(rst_jt_l)\delta_{ki} + \theta(rst_jt_k)\delta_{li}. \end{aligned}$$

It descends through the triangles and is an alternating CE cocycle, by the top-cover calculation of Theorem 3.1. That displayed construction also works at $d = 3$: completeness of the top cohomology classification in rank three is not needed here. It satisfies

$$\alpha_I(T_i, x_j) = \delta_{ij}, \qquad \alpha_I(\mathcal{S}_d, U) = 0.$$

Its ordinary Engel obstruction vanishes in rank three as well. In the notation $x = u + v, y = s + w$ with $u, s \in U$ and $v, w \in P_d$, the four obstruction terms are

$$\eta(t_u c(s, u), u) + \eta(t_u t_s v, u) + \beta(t_u c(s, u), v) + \beta(t_u t_s v, v).$$

The first has input weight four, below $d + 2$. The middle terms cancel by balance and $t_u^2 = 0$; the last is zero by the four-term formula. Thus the cocycle is three-Engel admissible for every $d \geq 3$.

C.3 Optimal packing

Put $p = \lfloor d/2 \rfloor$. Orient the complete graph so every outdegree is at most p: use the cyclic regular tournament for odd d, and delete one vertex of the $(d+1)$-vertex tournament for even d. At each vertex label its outgoing edges distinctly by $1, \dots, p$. In $K = k^p \oplus k e_{\mathrm{top}}$ use the cocycle

$$\alpha = \sum_{i \to j} e_{\mathrm{label}(i,j)} \zeta_{ij}^{i} + e_{\mathrm{top}} \alpha_I.$$

Its center-detection map is injective. The (i, label) coordinate detects one specific basis vector of $\mathcal{S}_d$; the final channel detects all of $\mathcal{T}_d$ by the identity matrix. Consequently the extension E has $Z(E) = K$ and attains the bound.

Every pair channel occurs in E'' as a bracket of c_{ij} with $t_i c_{ij}$ for an edge of that label. The top channel occurs as the bracket of c_{ij} with $t_{[d]\setminus\{j,k\}} c_{ik}$, for distinct i, j, k. Its pair pullbacks vanish. Hence $E'' = K$. Lifts of the d generators generate E: their generated algebra B has $E = B + K$, and centrality together with $K = E''$ forces $K \subset B$. The top pairing proves exact class $d + 2$.

The complete E_d maps onto this E over H_d. Its center must therefore have zero image in H_d. Since E_d'' is already central, $Z(E_d) = E_d''$ and $E_d / Z(E_d) = H_d$, proving Theorem 5.1. The graph orientation and the center-to-Hom injection are classical; the attaining cocycle gives the exact simultaneous center cost in this family.

D Stable central depths and cubic critical layers

This appendix proves Theorems 4.3 and 4.4. Put $M_0(H) = H''$, $M_{i+1}(H) = [M_i(H), H]$, and use the balanced module B_d of Appendix B.

D.1 The common coefficient presentation

All ordinary results below hold over every characteristic-two field k with $|k| > 2$. Expand $[v, u, u, u]$ in independent scalar parameters of arbitrary Lie values. Each parameter from u has degree at most three and each parameter from v degree at most one. Evaluations over k recover every coefficient, since $|k| \geq 4$. Thus the free ordinary three-Engel algebra is the base change of the presentation over $\mathbb{F}_2$ imposing all these formal coefficients. The central-depth identities are multilinear and commute with the same base change. In particular the finite relation matrices below, whose entries are zero and one, have the same ranks over every such k. This argument does not identify the coefficient-linearized variety over $\mathbb{F}_2$ with the possibly larger ordinary three-Engel variety there.

D.2 A general peeling and propagation principle

The peeling map (21) holds for every $r \geq 1$ when $M_{r+1} = 0$ and the balanced source degree is at least six. Indeed $[M_i, H'] \subset M_{i+2}$ makes changes of either derived lift or an outer lift vanish in M_{r+1}. The partial Engel identity with inputs a, a, v exchanges a coefficient action with the first outer input. A triangle is respected by peeling from its common coefficient, which is nonempty in that degree range. This establishes the map without introducing an action of R on the target quotient.

Suppose $s \geq 5$ and

$$\mathrm{cl}\, D_{r,s-1} = r + 2 \qquad (r \geq a). \tag{35}$$

Then $\mathrm{cl}\, D_{d,s} \leq d + 3$ for $d \geq a$. For $d \geq a + 1$, put $T_d = (D_{d,s})_{d+3}$. Every weight of T_d has full support: at most $d - 1$ inputs generate an algebra of class at most $d + 2$. Peeling gives

$$(B_d)_p \otimes U^{\otimes(s-2)} \twoheadrightarrow T_d, \qquad p = d + 5 - s. \tag{36}$$

Lemma D.1 (Propagation from the actual balanced source)**.** *If $d \geq \max(a+1, s+2)$, then $T_{d-1} = 0$ implies $T_d = 0$.*

Proof. Omit a coordinate y and substitute $x_i \mapsto x_i + \lambda[x_i, y]$ in an old critical identity. On the two heads the first variation is

$$\delta_{i,y} c_{ij} = t_y c_{ij} + t_i c_{yj}.$$

It extends by the product rule; coefficient actions are unchanged modulo H'', and varying an outer input raises the depth by an extra unit and vanishes. Thus the span S of these source suspensions has zero image in T_d. It is $\mathrm{GL}(U)$-stable. One may extract the coefficient either from the resulting polynomial, or by projecting to the degree increased by exactly one; all higher increments are above the central class bound $d+3$.

Fix a multilinear source on a p-element set I and a coordinate outer tail. If their combined support is deficient, its image is zero. Otherwise the tail covers all $d-p = s-5$ indices outside I and uses at most three indices of I. Choose $y \in I$ absent from the tail. A source symbol is a matching of two disjoint heads, with the other indices in its coefficient. If the matching omits y, suspending each endpoint of one head and adding the results gives the original matching, by the triangle on that head and y. If it contains y, replace that endpoint by a spare vertex; suspension relates it to a matching omitting y. Thus every multilinear source with the fixed tail lies in S. A spare vertex exists because $p \geq 7$. The old source has degree $p-1 \geq 6$, so both peeling maps are in range.

Every degree-p presentation monomial is in the $\mathrm{GL}(U)$-span of multilinear monomials. Replace one occurrence of a repeated variable x by an unused y. The invertible shear $y \mapsto y+x$ sends the new monomial to its sum with the old one. Induction on support deficit proves the assertion; no factorial is divided out. Since $d \geq p$, enough unused variables exist. Transport an outer tensor by the inverse shear before applying this argument, so no moved tail is incorrectly held fixed. This proves that the whole source has zero image, and hence $T_d = 0$. □

This proof uses the module before Koszul multiplication. A Jacobian element can have zero first variation and still belong to the span of variations of other elements. Its individual invariance is therefore not an obstruction to this source-level propagation.

D.3 Full support, singleton symmetry, and three profiles

The weights of a full-support degree-$d+3$ component are exactly

$$(4, 1, \dots, 1), \qquad (3, 2, 1, \dots, 1), \qquad (2, 2, 2, 1, \dots, 1).$$

If two variables occur once, shearing one into the other adds a term with deficient support, which is zero. The opposite shear also fixes the component, and three such shears give their transposition in characteristic two. All singleton permutations consequently act trivially.

Choose a singleton y. The component linear in y is the cyclic module for the associative envelope

$$A_3(H) = U(H)/\langle u^3 : u \in H \rangle$$

with the appropriate depth relations. To verify the universal statement, the semidirect bracket of an old algebra and a right module gives

$$[v+n, u+m, u+m, u+m] = nu^3 + m(u^2 v + uvu + vu^2).$$

The second coefficient is the partial polarization of the first. The free component linear in y is obtained by discarding terms with two y's, so this criterion is both necessary and sufficient there. Writing V_j for its part in M_j, one obtains

$$V_0 = (A_3(H)H)H', \qquad V_{j+1} = V_j H + A_3(H) M_j(H).$$

Both the all-factor recurrence and the equivalent ideal-generator recurrence are provided in the verification package.

To encode the other m singleton variables, use $R_m = k[t_1, \ldots, t_m]/(t_i^2)$ and replace them by $t_i Z$. An anonymous word of Z-degree $|S|$ carries coefficient t_S. The letter Z itself is not required to have cube zero. Impose cubes and repeated-value cubic coefficients only when the repeated Lie value has Z-degree zero; impose every distinct-value six-term coefficient. Repeating a positive-degree coefficient gives $t_S^2 = 0$. Distinct disjoint sets carrying the same anonymous value give even multiplicities and cancel.

This encoding preserves the complete critical component. Indeed, identifying two orders of singleton labels in a subword differs by a global singleton permutation fixing its surrounding context. If the context repeats a label the square-free product is already zero; otherwise that global permutation acts trivially as proved above. Conversely every anonymous Lie word can be labeled on any disjoint sets of the prescribed sizes. Hence the surviving cubic and depth relations are precisely those induced from the labeled component. No sum over $m!$ labelings is used. The three required old caps at rank d are therefore

$$(4, d-2), \qquad (3, 2, d-3), \qquad (2, 2, 2, d-4),$$

where the last entry counts singleton labels encoded by Z.

D.4 A cubic representation and an exterior-only alternative

The common coefficient presentation gives T_d an algebraic $\mathrm{GL}(U)$ action, with its literal integral multigrading. Full support implies that $\det(U)^{-1} \otimes T_d$ has nonnegative weights of total degree three. The classical weight criterion for polynomial representations makes it a homogeneous cubic representation [4, Section 12]. Finite-field diagonal characters alone would not establish this integral-weight statement.

If the repeated residual weight $(2, 1)$ is zero, the pure residual weight is zero too. Directly in the original $(4, 1, \ldots)$ weight, a root substitution has only a possible cubic coefficient: the linear and quadratic coefficients have the vanishing repeated shape, and the fourth has deficient support. The root-group law forces this cubic coefficient to vanish, because $(s+t)^3 + s^3 + t^3 = s^2 t + s t^2$ is not the zero polynomial. The opposite root fixes the vector, so their threefold product swaps its pure weight to a different weight and forces the vector itself to vanish.

Only the distinct cubic weights remain. Roots between two occupied indices fix each weight space, and so do roots between two unoccupied indices. Three shears show that the stabilizer permutations within either set act trivially. A root moving an occupied index to an unoccupied one has just its linear surviving coefficient. The opposite-root identity identifies this coefficient with the permutation identification between the two weight spaces. Thus

$$T_d \simeq \det(U) \otimes \Lambda^3(U) \otimes E \tag{37}$$

for a trivial multiplicity space E. This proof also shows directly, without the polynomial criterion, that permutations of the three doubled anchors act trivially whenever the repeated profile vanishes. It justifies quotienting by those permutations at the final degree; they are not imposed in lower degrees.

D.5 Exact finite starts and the stable ranges

At depth five and rank nine the three anonymous old caps are $(4, 7), (3, 2, 6), (2, 2, 2, 5)$. Complete relation elimination gives

cap	$(4, 7)$	$(3, 2, 6)$	$(2, 2, 2, 5)$
word dimension	330	4620	41580
relation rank	330	4620	41580

at old degree eleven. Hence $T_9 = 0$. Reverse-coordinate replays give the same ranks. The first two also complete the all-factor depth recurrence; the third uses the ideal-generator recurrence, with the all-factor control through the preceding degree. These are exact matrices, not sampled evaluations. Lemma D.1, with $s = 5, a = 6$, proves $\operatorname{cl} D_{d,5} \le d + 2$ for every $d \ge 9$.

At depth six and rank ten the repeated cap $(3, 2, 7)$ has dimension zero. Both recurrences and reverse coordinates agree in every grade. The exterior alternative kills the pure cap and permits the top anchor-permutation quotient in $(2, 2, 2, 6)$. Its 83160 coordinates reduce to 13860 orbits. The resulting relation rank is 13860. The preceding old-degree-eleven component has dimension 31. Two complete eliminations, one using lower-multidegree equivariance and one using reversed coordinates with no such replication, agree in every grade. Thus $T_{10} = 0$. Apply Lemma D.1 with $s = 6, a = 9$. The center-by-metabelian covers attain $d + 2$ in both stable ranges. This proves Theorem 4.3.

The computations share elementary word arithmetic; they are different coordinate eliminations rather than wholly unrelated implementations. The source files, ranks, and literal lower models are included in `verification/stable_depth/`, with separate replay commands.

D.6 The two boundary representations

The complete critical weight dimensions in ranks seven and eight, ordered as pure, repeated and distinct, are respectively

$$(0, 1, 2), \qquad (1, 1, 1).$$

Their total dimensions are $7 \cdot 6 + 2\binom{7}{3} = 112$ and $8 + 8 \cdot 7 + \binom{8}{3} = 120$. For rank seven choose a nonzero repeated weight vector. Two polarizations give the independent distinct-weight vectors corresponding to $[[a, b], c]$ and $[[a, c], b]$. For rank eight each weight is a line. Singleton symmetry and the trivial one-dimensional characteristic-two character of the remaining anchor stabilizer make the transported weight bases well defined.

The explicit bracket words and all root coefficients are recorded in `check_rank7_cubic_module_actions.py` and `check_rank8_cubic_module_actions.py`. The former checks 45 root coefficients and 2160 permutation cases against the free-associative realization of $\mathrm{Lie}^3(U)$; the latter checks 54 root coefficients against $\mathrm{Sym}^3(U)$. Diagonals have the displayed multidegrees. Choose an unaffected singleton for y in each check; singleton symmetry makes the choice irrelevant. Roots, diagonals and permutations generate $\mathrm{GL}(U)$, so these intertwining maps and the equal dimensions prove Theorem 4.4. This uses the actual action, not just a coincidence of dimensions.

The usual characteristic-two sequence

$$0 \longrightarrow U^{[1]} \otimes U \longrightarrow \mathrm{Sym}^3 U \longrightarrow \Lambda^3 U \longrightarrow 0, \qquad a^{[1]} \otimes b \longmapsto a^2 b,$$

has repeated-monomial kernel of dimension d^2. It does not split for $d \ge 3$: a section on the distinct weight would send $e_1 \wedge e_2 \wedge e_3$ to $e_1 e_2 e_3$, but the shear $e_3 \mapsto e_3 + e_1$ fixes the former and changes the latter by $e_1^2 e_2$. Thus the rank-eight critical layer has a nonsplit submodule of dimension 64 and quotient of dimension 56. The cubic functors and this sequence are classical representation theory; their occurrence here identifies the two Engel boundary layers.

D.7 Literal boundary models and the complete bottom

The rank-seven and rank-eight lower models have dimensions 294 and 586 over $\mathbb{F}_4$, with operator/module decompositions $144 + 150$ and $288 + 298$. They are graded and generated by seven and eight degree-one elements. Their lower-central dimensions are

$$(294, 287, 276, 234, 164, 109, 59, 17, 3, 1, 0),$$
$$(586, 578, 565, 509, 397, 257, 187, 75, 19, 3, 1, 0),$$

and their second-derived series dimensions are $(94, 38, 10, 2, 1, 0)$ and $(187, 75, 19, 3, 1, 0)$. The tables and separately implemented verifier check Jacobi, all cubic coefficients over $\mathbb{F}_4$, generation and explicit nonzero words. They prove the lower classes 10 and 11; the central-step upper bound proves equality. Rank eight therefore excludes a stable fifth-depth start below nine.

For either lower model B, form the marked generated fiber product of B with E_d over $B/M_1(B)$. Write it as C. For $j \geq 1$,

$$M_j(C) = M_j(B) \times 0, \qquad C/M_1(C) = E_d.$$

The first assertion follows from projection onto B and the vanishing of $M_1(E_d)$; the second follows from the common quotient. Hence the quotients $C/M_j(C)$ have successive classes $d+2, d+2, d+2, d+2, d+3$. This proves the ordinary attaining chains over the complete bottom.

D.8 A classical comparison locating the next obstruction

For the ordinary current algebra $H_m = kx \ltimes (S \otimes A_m)$ from Appendix E, write $U_{\geq r}$ for supports of size at least r, and likewise V, W. Direct brackets give

$$M_{2j} = U_{\geq j+2} + V_{\geq j+3} + W_{\geq j+3}, \qquad M_{2j+1} = U_{\geq j+3} + V_{\geq j+3} + W_{\geq j+4}.$$

The weights $2|I|+1, 2|I|, 2|I|-1$ therefore yield

$$\operatorname{cl}(H_m/M_s) = \min(2m+1, s+4). \tag{38}$$

Its direct fifth-depth truncations have class at most nine, so they do not supply the rank-seven and rank-eight boundary models above. For every $s \geq 3$, taking $m = s$ gives a rank-$(s+1)$ object of class $s+4$. Any eventual stable starting rank a_s for the law $\operatorname{cl} D_{d,s} = d+2$ must consequently satisfy $a_s \geq s+2$. These are consequences of the classical current data, distinct from the critical-layer vanishing used for the new stable ranges.

E The ordinary current comparison

Put $R_m = k[t_1, \ldots, t_m]/(t_1^2, \ldots, t_m^2)$ and $A_m = (t_1, \ldots, t_m)$. For nonempty $B \subseteq [m]$, use the square-free monomials t_B. Let S have basis U, V, W and table $[U, V] = U$, $[U, W] = V$, $[V, W] = W$. The derivation $\delta = \operatorname{ad} U$ sends W to V, V to U, and U to zero. Define $H_m = kx \ltimes (S \otimes A_m)$, with x acting by δ. This is the classical disjoint-support current construction [7, Proposition 2.1], with the standard derivation described in [6, Proposition 3.3(1)].

Writing $U_B = U \otimes t_B$, and similarly for V, W, the brackets are the displayed S-table on disjoint supports, zero on overlapping supports, and $[x, W_B] = V_B$, $[x, V_B] = U_B$. This proves Jacobi by the semidirect construction. To check three-Engel, write an element as (a, u, v, w) and put $h = a + u$. Its adjoint on the current ideal and its square have matrices

$$T = \begin{pmatrix} v & h & 0 \\ w & 0 & h \\ 0 & w & v \end{pmatrix}, \qquad T^2 = \begin{pmatrix} hw & hv & a^2 \\ vw & 0 & hv \\ 0 & vw & hw \end{pmatrix}.$$

The identities $u^2 = v^2 = w^2 = 0$ give $T^3 = 0$. The image of an external element is a multiple of $(v, w, 0)$, which T^2 kills. Thus H_m is three-Engel over every field under consideration. The generators $x, W_{\{i\}}$ give exact ordinary rank $m+1$. Weights $1, 2|B|+1, 2|B|, 2|B|-1$ on x, U_B, V_B, W_B add under brackets. The word $[W_{\{1\}}, x, x, W_{\{2\}}, x, \ldots, W_{\{m\}}, x] = U_{[m]}$ therefore proves exact class $2m+1$.

For support cutoffs write $U_{\geq j}$ for the span of the U_B with $|B| \geq j$. Direct bracketing gives $H''_m = U_{\geq 2} + V_{\geq 3} + W_{\geq 3}$. Bracketing once exchanges this with $U_{\geq 3} + V_{\geq 3} + W_{\geq 4}$; each subsequent pair of brackets increases all three cutoffs by one. This proves the recurrences used in (38). In each quotient the largest surviving weight is achieved by the same disjoint-support words, proving its exact class, rather than only a grading bound.

F Exact starting systems and reproducibility

The finite starting systems in the central-depth proofs are supplied with their literal inputs and replay programs. The collecting seed checker expands the 22-node identity in free associative words, checks Lie membership of substituted polynomials and tests damaged certificates. The rank-six controls replay every imposed degree and multidegree through weight nine for the three capped profiles. They verify full relation ranks and that the new bracket coordinates span the entire universal kernel. The symbolic suspension and support arguments, rather than rank extrapolation, prove the general rank assertions.

The fifth-depth boundary tables have separate verifiers for Jacobi, all cubic coefficients, generator closure and filtrations. Cubic-module controls compare actual generator substitutions. For the sixth-depth last system, the supplied independent replay reverses coordinates and does not use the lower-multidegree replication optimization; its final rank is 13860 after the proved top anchor quotient. The exact ordinary minimum-center constructions are also reconstructed.

All necessary tables and local helpers are in the independent `verification/` directory. Run

```
python verification/run_replay.py --output-dir replay_core
```

and add `--extended` for the complete expensive stable-start systems. A fresh report records which tasks were rerun; inherited reports retain their original scope. The runner preserves all input files and collects failures. The controls use the Python standard library. Finite proof inputs, symbolic propagation and model checks have separate roles; no proof-assistant formalization is claimed.

References

[1] L. G. Kovács and R. Stöhr, Free centre-by-metabelian Lie algebras in characteristic 2, *Bulletin of the London Mathematical Society* **46** (2014), no. 3, 491–502. doi:10.1112/blms/bdu011. Author manuscript: `https://archives.maths.anu.edu.au/people/Kovacs/K114.pdf`.

[2] Yu. V. Kuz'min, Free center-by-metabelian groups, Lie algebras, and D-groups, *Math. USSR-Izv.* **11** (1977), no. 1, 1–30. doi:10.1070/IM1977v011n01ABEH001443.

[3] N. Mansuroğlu and R. Stöhr, Free centre-by-metabelian Lie rings, *The Quarterly Journal of Mathematics* **65** (2014), 555–579. doi:10.1093/qmath/hat017.

[4] O. Mathieu, Tilting modules and their applications, in *Analysis on Homogeneous Spaces and Representation Theory of Lie Groups*, Advanced Studies in Pure Mathematics **26** (2000), 145–212. Section 12. `https://www.imsc.res.in/~knr/vdk/mathieu.pdf`.

[5] Ş. Papadima and A. I. Suciu, Chen Lie algebras, *International Mathematics Research Notices* **2004** (2004), no. 21, 1057–1086. doi:10.1155/S1073792804132017.

[6] H. Strade, *Lie algebras of small dimension*, preprint, 2006. `https://arxiv.org/abs/math/0601413`.

[7] G. Traustason, Engel Lie-algebras, *The Quarterly Journal of Mathematics* **44** (1993), no. 3, 355–384. doi:10.1093/qmath/44.3.355.

[8] G. Traustason, A polynomial upper bound for the nilpotency classes of Engel-3 Lie algebras over a field of characteristic 2, *Journal of the London Mathematical Society* **51** (1995), no. 3, 453–460. doi:10.1112/jlms/51.3.453.

[9] B. Li, *Quadratic orbit maps and three-Engel Lie algebras in characteristic two*, companion manuscript, 2026.

[10] B. Li, *Extremal dimensions and critical isomorphism growth of metabelian three-Engel Lie algebras*, companion manuscript, 2026.

[11] B. Li, *Quadratic maps satisfying $f \circ f = f + \mathrm{id}$: nilpotence, Lie reconstruction, and growth*, unified research manuscript, 2026. Earlier public version: doi:10.5281/zenodo.23852642; expanded local revision, 2 November 2026.